

Matthew Cao

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EDUCATION

University of California, Berkeley (UC Berkeley)

May 2026.

B.A. Molecular & Cell Biology - Molecular Therapeutics

Berkeley, CA

B.A. Data Science - Computational Methods in Molecular & Genomic Biology

EXPERIENCE

Institute on Aging - Cognitive Engagement and Healthy Aging Program

San Francisco, CA

Cognitive Engagement Volunteer

Oct. 2025 – May 2026

- Led individual/group patient-engagement sessions for 30+ older adults, building rapport and adapting communication to cognitive and physical abilities.
- Delivered individualized cognitive and daily-living activities—including memory games, language practice, visual communication, eating, and dressing support—for participants with stroke, aphasia, and cognitive decline.
- Documented attendance, communicated patient observations to staff, and adhered to facility safety and dietary protocols.

UCSF Genitourinary Oncology - Precision Cancer Medicine and Clinical Oncology

San Francisco, CA

Clinical Shadowing

Jan. 2026 – May 2026

- Shadowed visits with clinical-trial patients in genitourinary oncology, including eligibility screening, lab/imaging review, adverse-effect monitoring, and informed-consent discussions.
- Supported Dr. Zhu in clarifying a patient's contrast-reaction concern; observed coordination with the RN and technicians to uphold consent requirements before testing.

UCSF Oncology Database - Oncology Clinical Trials Research

Remote

Research Intern

Sep. 2025 – Aug. 2026

- Produced standardized clinical-trial records used by medical oncologists to evaluate emerging oncology therapies, treatment evidence, and general biostatistics.
- Abstracted eligibility criteria, study design, treatment arms, endpoints, and efficacy and safety outcomes from publications and federal trial registries for 50+ oncology clinical trials.
- Resolved conflicting data through verification and supervisor escalation; completed QC review before database publication.

Gracell Biotech./AstraZeneca - LNP in vivo CAR-T Development

Shanghai, China

Research Apprentice

May 2024 – Aug. 2024

- Executed SOP-driven LNP formulation, purification, and optimization workflows, followed by in vitro and in vivo testing to support preclinical CAR-T development.
- Maintained complete, contemporaneous experimental records—including formulation details, production dates and times, and test conditions—to support traceability, reproducibility, and data quality.

Junior Care Clinic - Community Health Advocacy and Outreach

Berkeley, CA

Vice President of External Affairs

Aug. 2024 – Jun. 2026

- Directed a cross-functional health education initiative involving 60+ contributors across 5 organizational committees, managing project development, timelines, and inter-team coordination from inception to publication of 25+ articles.
- Led final quality review, verifying sources and editing health-education materials for East Bay communities.

Liu Lab - Studies of Nucleic Acid-based Agents for Inhibition of Viral Replication

Berkeley, CA

Research Apprentice

Jan. 2024 – Jan. 2025

- Built 3D protein models with AlphaFold 2/3 and ChimeraX to evaluate vaccine development strategies.
- Annotated genomes of SARS-CoV-2 and SARS-CoV-229E, identifying key transcriptomic features.
- Analyzed and generated 3D structures of the SARS-CoV-2 envelope protein.

University of California, Berkeley

Berkeley, CA

Department of Chemistry Exam Proctor

Jan. 2023 – May 2026

- Administered university examinations while ensuring compliance with testing protocols and academic integrity standards.
- Coordinated examination logistics and maintained accurate student records in a high-volume academic environment.

Suitcase Clinic - Healthcare Access for Unhoused and Underserved Communities

Berkeley, CA

Community Health Volunteer

Jan. 2025 – Jan. 2026

- Assisted 20+ underserved community members in accessing healthcare, social, and community resources.
- Worked with interdisciplinary volunteer teams to support client needs during outreach events.
- Communicated with individuals from diverse backgrounds to assess needs and connect them with appropriate services.

RESEARCH PROJECTS

RNA-Splicing Model Interpretability with Synthetic Transcriptomes

Lareau Lab, Innovative Genomics Institute

Independent Research Project Manuscript

Jul. 2025 – Dec. 2025

- Developed Python workflows using genomic databases and annotated splice-site data to test how sequence-feature disruption affected RNA-splicing model predictions.
- Generated synthetic transcriptomes with selectively altered coding-sequence composition and splice-regulatory motifs, then compared model outputs before and after perturbation.
- Found that substantial sequence-feature disruptions produced limited prediction changes, narrowing simple mechanistic explanations and raising questions about learned representations in genomic deep-learning models.

Benchmarking cNMF Methods for Single-Cell Gene-Program Inference

UC Berkeley

CS 194/294 Computational Biology Project Manuscript

Dec. 2025

- Benchmarked Standard cNMF and Poisson cNMF against a linear scVI/LDVAE deep-learning baseline for single-cell analysis using PBMC 3k and Paul15 datasets.
- Contributed to the abstract, figures, results, discussion, formatting, and auxiliary manuscript sections, with emphasis on interpreting model agreement and biological coherence.
- Helped evaluate whether inferred gene programs mapped to significant Gene Ontology processes using top-gene enrichment, supporting the conclusion that cNMF methods produced more biologically interpretable programs than the deep-learning baseline in the tested settings.

Regional Differences in Heart Disease Modeling

UC Berkeley

Project Lead, IEOR 142A Data Science Project Manuscript

May 2026

- Developed project codebase analyzing UCI Heart Disease datasets from Cleveland, Hungary, Switzerland, and VA medical-center populations.
- Used PCA and K-means clustering to show stronger separation by geographic origin than by disease severity.
- Modeled binary heart-disease presence with regularized logistic regression and bootstrap confidence intervals.

SKILLS & CERTIFICATIONS

Basic Life Support Certified (Credential ID 265411388250)

Data Analysis & Visualization: Excel, SQL, Python (NumPy, Matplotlib, SciPy), MATLAB

Office & Productivity Tools: Google Workspace (Docs, Sheets, Slides), Microsoft Office (Word, Excel, PowerPoint)

Laboratory Techniques: PCR, ELISA, FACS, Cell Culture, Chromatography

Design & Graphics: Website Design & Maintenance (HTML, CSS, JS), Adobe Suite (Photoshop, Illustrator)

Imaging & Computational Modeling: AlphaFold 2/3, ChimeraX, SpliceAI, BLAST, FASTQC